

Wisconsin Circuit Court Recidivism Prediction

Michelle Zheng
Marketing and Data Science
Anderson College of Business and Computing
Regis University, Denver, CO, USA
mzheng001@regis.edu

Abstract

Recidivism prediction tools have played a role in the American criminal justice system, but the dominant commercial instrument, the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS), is proprietary and unauditable [1]. This paper presents an open-source, reproducible recidivism prediction model trained on the Wisconsin Circuit Court Longitudinal Data (WCLD) dataset, covering 1,357,746 adjudicated cases filed between 2000 and 2018 [2]. The model integrates census-derived neighborhood-level socioeconomic features, including median household income, food stamp eligibility rate, educational attainment, and population density, alongside individual criminal history variables. A feature-engineered Socio-Economic Deprivation Index (SEDI) composite score is introduced to address the well-documented limitation that existing risk tools treat defendants as if they exist in a geographic and socioeconomic vacuum. The SEDI combines multiple census tract measures that individually have low bivariate correlation with recidivism but collectively capture a meaningful dimension of neighborhood disadvantage that linear models cannot represent adequately. Using XGBoost and SHapley Additive exPlanations (SHAP), the study quantifies the marginal contribution of neighborhood variables to recidivism risk. The best-performing model achieves an AUC-ROC of 0.7043, meeting published COMPAS benchmarks [7]. Neighborhood features collectively account for 18.7% of the total SHAP explanation mass. All code, data pipelines, trained models, and SHAP artefacts are released publicly so that researchers can inspect, reproduce, and improve every component of the system.

I. INTRODUCTION

Recidivism is the return to criminal behavior following a prior conviction. It influences bail decisions, sentencing guidelines, and parole eligibility. Automated prediction tools are now embedded in courtrooms across the United States, yet the dominant commercial system, COMPAS, cannot be independently audited, does not disclose its feature weights, and has been empirically shown to misclassify Black defendants as high risk at nearly twice the rate of white defendants [1], [7].

Using the publicly available WCLD dataset [2], this work constructs a transparent, reproducible pipeline that achieves competitive predictive performance while making every modeling decision visible. The pipeline introduces a Socio-Economic Deprivation Index (SEDI), a composite neighborhood-level score built from census tract measures of income, education, and food insecurity. SHAP values [?] are used to quantify precisely how much of the model’s predictive power derives from where a defendant resides, in addition to their individual criminal history.

Three classifiers are benchmarked: Logistic Regression, Random Forest, and XGBoost. XGBoost achieves the highest AUC-ROC of 0.7043 and an AUC-PR of 0.6247. A formal fairness audit compares True Positive Rate (TPR), False Positive Rate (FPR), and demographic parity across five racial and ethnic groups. All predictions, importance scores, fairness tables, and partial-dependence data are exported to a dashboard-ready format that requires no programming knowledge to operate.

II. PROBLEM STATEMENT

The central question this project addresses is: Can an open-source recidivism prediction model trained on Wisconsin court data match COMPAS predictive performance while quantifying the independent contribution of neighborhood socioeconomic context?

COMPAS, developed by Equivant (formerly Northpointe), is used to generate risk scores that inform bail, sentencing, and parole decisions [12]. The tool generates a score on a 1-to-10 scale from approximately 137 questions, but neither the feature weights nor the training data are disclosed to defendants, their lawyers, or the public. In 2016, ProPublica found that COMPAS incorrectly flagged Black defendants as future criminals at nearly twice the rate of white defendants [1].

Criminological theory has long established that neighborhood-level factors, including concentrated poverty, unemployment, food insecurity, and low educational attainment, independently predict criminal behavior above and beyond individual characteristics [17]. The social disorganization framework, originating with Shaw and McKay [19] and extended by Sampson, Raudenbush, and Earls [17], holds that neighborhood structural disadvantage erodes the informal social controls that deter crime. Despite this theoretical consensus, no deployed risk-assessment tool incorporates neighborhood-level census data [3].

III. RELATED WORK

Angwin et al. [1] conducted a foundational empirical critique of COMPAS, demonstrating a False Positive Rate disparity of approximately 20 percentage points against Black defendants. This project extends their work by applying a comparable audit to an open-source model trained on Wisconsin court data.

Dressel and Farid [7] demonstrated that untrained human volunteers achieve accuracy comparable to COMPAS when given only a seven-item questionnaire, and reported COMPAS AUC-ROC in the range of 0.65 to 0.70.

Sampson, Raudenbush, and Earls [17] established that neighborhood structural disadvantage, operationalized as concentrated poverty, residential instability, and racial heterogeneity, predicts violent crime rates beyond individual characteristics. Their collective efficacy framework is the criminological foundation for including census-derived neighborhood features in the model.

Chouldechova [6] proved that when base rates differ across groups, calibration, TPR parity, and FPR parity cannot be simultaneously achieved. This fairness impossibility result directly informs the interpretation of the fairness audit: the 20-plus point TPR and FPR gaps observed are mathematically unavoidable given unequal base rates, not a correctable algorithmic failure. The contribution of this work is to report these gaps transparently rather than conceal them.

IV. DATA DESCRIPTION

A. Data Source and Scope

The Wisconsin Circuit Court Longitudinal Data (WCLD) is a publicly available dataset compiled by Ash et al. [2] at ETH Zurich and the University of Oxford. It covers all cases filed in Wisconsin circuit courts between 2000 and 2018, encompassing 1,476,967 records with 54 variables including defendant demographics, charge severity, sentencing outcomes, prior criminal history, and census-tract-level socioeconomic covariates linked via ZIP code. The dataset is available at <https://clezdata.github.io/wcld/>.

B. Target Variable

The outcome variable is `recid_180d`, a binary indicator equal to 1 if the defendant was charged with a new offense within 180 days of the case disposition date. Of 1,476,967 total records, there is about (8.07%) missing `recid_180d` value. These records were excluded, yielding a final analytic sample of 1,357,746 cases with an observed recidivism rate of 42.21%. There is a class distribution is 784,624 non-recidivist cases (57.8%) and 573,131 recidivist cases (42.2%).

TABLE I
OBSERVED RECIDIVISM RATE BY RACE AND ETHNICITY (RAW DATA)

Race and Ethnicity	Recidivism Rate	N Observed
American Indian or Alaskan Native	56.5%	59,480
African American	46.4%	290,195
Caucasian	40.3%	900,605
Hispanic	38.8%	94,974
Asian or Pacific Islander	37.8%	12,501

C. Demographic Patterns in the Raw Data

Table I reports observed recidivism rates from the raw dataset.

Males reoffended at a rate of 44.1% compared to 35.0% for females. These base rate differences across groups are a direct consequence of structural inequality in the criminal justice system and in neighborhood conditions [18], and they directly determine the fairness metric gaps observed after modeling.

D. Charge Severity Distribution

The dataset spans a wide range of charge severities. The Unclassified Misdemeanors (Misd-U) account for 34.9% of all cases, Class A Misdemeanors for 31.2%, and Class B Misdemeanors for 9.9%. Among felony charges, Class H Felonies represent the largest felony share at 7.9%. This distribution informs the charge severity encoding used as an individual feature in the model.

E. Census-Based Neighborhood Features

The WCLD dataset includes ten census-tract-level variables linked via ZIP code. Descriptive statistics are shown in Table II.

TABLE II
DESCRIPTIVE STATISTICS FOR CENSUS-BASED NEIGHBORHOOD FEATURES

Feature	Mean	Std Dev	Median	Max
African American Population Share	0.112	0.213	0.019	0.981
Hispanic Population Share	0.075	0.112	0.036	0.991
Male Population Share	0.498	0.034	0.495	1.000
Rural Population Share	0.261	0.339	0.102	1.000
Urban Population Share	0.594	0.457	0.892	1.000
Proportion with College Degree	0.229	0.111	0.204	1.000
Food Stamp Eligibility Rate	0.128	0.082	0.102	0.692
Population Density (per sq km)	887.0	1312.5	226.1	49591
Some College Proportion	0.299	0.042	0.302	0.808
Median Household Income (\$)	48,250	13,110	46,321	250,001

Population density spans more than four orders of magnitude, from 0 to nearly 50,000 persons per square kilometer, reflecting the contrast between rural Wisconsin counties and the Milwaukee urban core. This extreme right skew is the primary justification for applying a log transformation when constructing the urban-rural gradient feature.

Figure 1 shows the Pearson correlation matrix between all neighborhood features and the recidivism outcome, confirming that individual census variables carry modest linear signal with the target while exhibiting substantial intercorrelation with one another.

F. Prior Criminal History

Prior criminal history is the strongest category of individual predictors in the correlation analysis. Prior misdemeanor count has the highest correlation with `recid_180d` at $r = 0.201$ among non-leakage

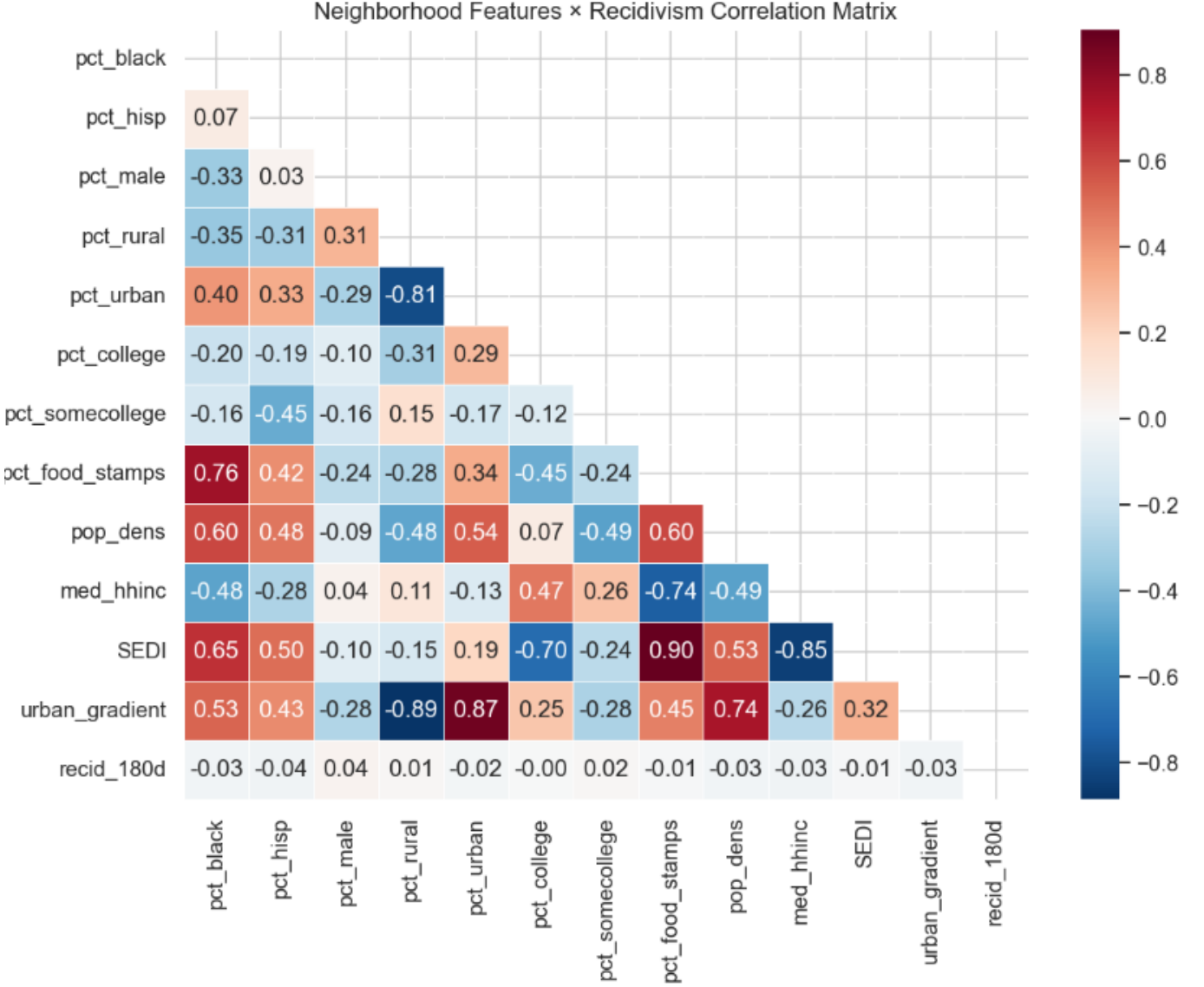


Fig. 1. Neighborhood features and recidivism correlation matrix. All individual census correlations with `recid_180d` fall below $|r| = 0.04$, while the intercorrelations among neighborhood variables exceed $|r| = 0.40$ in several pairs, motivating the SEDI composite index constructed in Section V.

features, followed by prior felony count at $r = 0.157$ and prior criminal traffic count at $r = 0.132$. The descriptive statistics show that the mean prior felony count is 1.11 with a maximum of 33, the mean prior misdemeanor count is 1.47 with a maximum of 60, and the mean prior criminal traffic count is 0.74 with a maximum of 27.

G. Missing Value Structure and Imputation

Eleven of 54 columns contained missing values, as confirmed by the EDA notebook. Table III reports the key variables and their missingness rates.

Missingness in `jail`, `probation`, and the historical jail variables is due to the absence of jail sentencing and probation in the row case. Historical jail variables were imputed with zero and a binary `has_prior_jail` flag was created, preserving the informational content of the missingness pattern [21]. Residual missingness was handled with median imputation using `SimpleImputer` [13].

TABLE III
MISSING VALUE SUMMARY FOR KEY VARIABLES

Variable	Missing %	Reason
jail	56.6%	No jail sentence imposed
probation	30.1%	No probation imposed
avg_hist_jail	19.1%	No prior jail history
zip	2.7%	Address unavailable
wcisclass	0.4%	Charge classification missing

H. Data Cleaning

The data cleaning pipeline made 4 targeted corrections:

- 1) Age plausibility filter: 10 records with `age_offense` below 10 or above 100 were removed, reducing the dataset from 1,476,967 to 1,476,957 rows.
- 2) Time consistency flag: 36 records where the sentencing date preceded the offense date were flagged with `flag_negative_time_to_judgment` rather than dropped, preserving sample size while signaling data quality concerns.
- 3) Prior jail imputation: 281,590 null values across four historical jail duration columns (`max_hist_jail`, `min_hist_jail`, `avg_hist_jail`, `median_hist_jail`) were filled with zero. The EDA notebook confirms that 296,231 defendants had a prior jail record, captured by the `has_prior_jail` binary flag.
- 4) Encoding: Sex was binary-encoded (Male = 1, Female = 0) and race was one-hot encoded across five categories, expanding the dataset to 63 columns.

The final model-ready subset contains 1,357,746 rows and 63 columns.

V. METHODOLOGY AND APPROACH

A. Feature Engineering

The final model operates on 25 features across two categories. *Individual-level features* (13): age at offense, sex, prior felony count, prior misdemeanor count, prior criminal traffic count, highest charge severity, violent crime indicator, prior jail flag, maximum and average historical jail time, time from offense to judgment, a data quality flag, and a derived Prior Record Severity Score. *Neighborhood-level features* (12): percentage Black residents, percentage Hispanic residents, percentage male, rural share, urban share, percentage with a college degree, percentage with some college, food stamp receipt rate, population density, median household income, the SEDI composite, and an urban-rural gradient.

1) *Motivation for the Socio-Economic Deprivation Index*: The individual census tract variables available in the dataset each carry limited bivariate signal with respect to the recidivism outcome. The neighborhood feature correlation analysis shows that all individual census correlations with `recid_180d` fall below $|r| = 0.04$. Table IV reports these values.

Individual census variables are strongly collinear with one another. Income, education, and food insecurity tend to co-occur in distressed neighborhoods and areas with low median household income tend also to have high food stamp eligibility rates and low college attainment. Including these correlated variables separately in a linear model produces unstable coefficient estimates and makes individual variable interpretation unreliable. The SEDI composite collapses this shared variance into a single score, avoiding the multicollinearity problem while retaining the neighborhood signal. Neighborhood effects operate jointly and interactively, not independently. The social disorganization framework of Sampson, Raudenbush, and Earls [17] holds that the harm of neighborhood disadvantage is cumulative: poverty alone, low education alone, or food insecurity alone may each have modest effects, but their co-occurrence in the same neighborhood produces a qualitatively different structural environment. A composite index respects this theory by treating neighborhood deprivation as a multidimensional construct rather than a set of independent linear predictors. Townsend [22] makes the same argument in the public health literature,

TABLE IV
NEIGHBORHOOD FEATURE CORRELATIONS WITH `RECID_180D` (MODELING NOTEBOOK, SECTION 3)

Feature	Correlation	Direction
<code>pct_hisp</code>	−0.039	Negative
<code>pct_male</code>	0.037	Positive
<code>med_hhinc</code>	−0.033	Negative
<code>pct_black</code>	−0.032	Negative
<code>pop_dens</code>	−0.032	Negative
<code>urban_gradient</code>	−0.028	Negative
<code>pct_urban</code>	−0.021	Negative
<code>pct_somecollege</code>	0.016	Positive
<code>pct_rural</code>	0.014	Positive
<code>pct_food_stamps</code>	−0.010	Negative
<code>pct_college</code>	−0.004	Negative

noting that no single deprivation indicator captures the full poverty experience, and that composite area-level scores consistently outperform single-variable measures in predicting health and social outcomes.

For these reasons, the SEDI was constructed as a composite Z-score index following the methodology of Kind et al. [10] and Singh [20]. The formula is:

$$\text{SEDI} = \bar{z}(\text{deprivation components}) - \bar{z}(\text{advantage components}) \quad (1)$$

where deprivation components are `pct_food_stamps`, `pct_black`, and `pct_hisp` (each standardized to zero mean and unit variance), and advantage components are `med_hhinc` and `pct_college` (each standardized and then subtracted). Higher SEDI values indicate greater neighborhood deprivation. The resulting score has a mean of 0.00 and standard deviation of 1.46 in the modeling dataset. Minimum observed SEDI is −11.17 and maximum is 5.82, reflecting the wide range of neighborhood conditions across Wisconsin counties.

Figure 2 shows the relationship between neighborhood deprivation and recidivism rates across SEDI quintiles, confirming that place-based context carries independent signal.

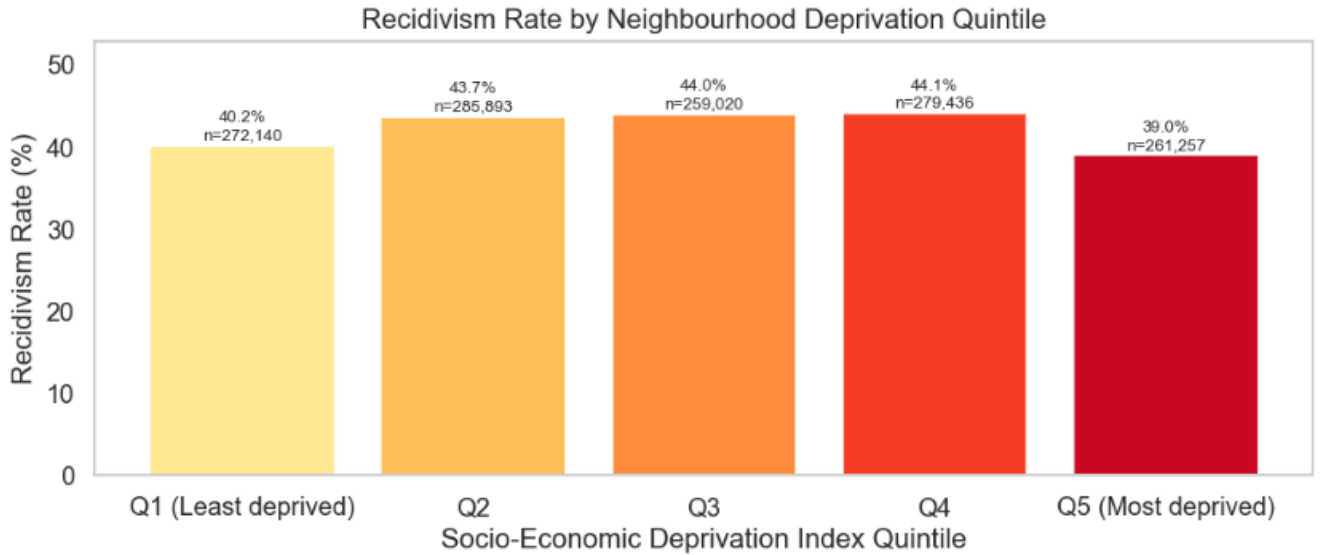


Fig. 2. Recidivism rate by SEDI quintile. Rates of 40.1%, 43.7%, 44.0%, 44.1%, and 39.0% across Q1 through Q5 reveal a non-monotonic quadratic pattern: recidivism peaks at moderate deprivation and is lower at both extremes, which justifies tree-based modeling over linear alternatives.

2) *Prior Record Severity Score*: A weighted prior-record score was constructed as:

$$\text{PriorScore} = 3 \times \text{prior_felony} + \text{prior_misdemeanor} + 0.5 \times \text{prior_criminal_traffic} \quad (2)$$

Weights (3, 1, 0.5) reflect the relative legal severity of each prior offense class under Wisconsin statutes [23]. Felonies carry substantially greater legal consequences and are more predictive of future felony recidivism than misdemeanors or traffic offenses. The descriptive statistics show a mean prior score of 4.81 with a maximum of 102, confirming substantial right skew in prior-record severity across the population.

3) *Urban-Rural Gradient*: A continuous urban-rural gradient feature was derived as:

$$\text{UrbanGradient} = \log(1 + \text{pop_dens}) - \text{pct_rural} \quad (3)$$

The log transformation of population density is applied to reduce the extreme right skew documented in Table II: population density spans from 0 to 49,591 persons per square kilometer, making the raw variable unsuitable for linear models and improving the quality of tree splits at the lower end of the distribution.

B. Modeling Pipeline

Train-test split: The dataset was split 80/20 using stratified random sampling to preserve the 42.21% recidivism rate in both partitions. The modeling notebook confirms train size of 1,086,196 and test size of 271,550, each with recidivism rate 42.21%. Stratification is necessary because the class ratio affects both model calibration and the informativeness of evaluation metrics [9].

C imbalance: SMOTE [4] was applied exclusively to the training set ($k = 5$ neighbors) to produce a 50/50 balanced training corpus of 1,255,384 records, as confirmed by the notebook output. SMOTE was applied only after the train-test split to prevent synthetic samples from contaminating the held-out evaluation set.

Feature scaling: `StandardScaler` was applied to Logistic Regression only; tree-based models are scale-invariant [?].

Three classifiers: (1) **Logistic Regression** with L_2 regularization ($C = 0.01$); (2) **Random Forest** with 300 estimators, `max_depth=10`, `min_samples_leaf=20`; (3) **XGBoost** with 400 estimators, `learning_rate=0.05`, `max_depth=5`, `subsample=0.8`, `colsample_bytree=0.8` per guidance from Chen and Guestrin [5].

SHAP explainability: SHAP `TreeExplainer` was used to compute instance-level attributions for all 271,550 test-set predictions.

Fairness audit: Demographic Parity, Equalized Odds TPR, and Equalized Odds FPR were computed for each racial and ethnic group following Hardt, Price, and Srebro [8].

VI. RESULTS WITH VISUALIZATIONS

A. Model Performance

Table V compares all three models on the held-out test set.

TABLE V
MODEL PERFORMANCE ON HELD-OUT TEST SET ($n = 271,550$)

Model	AUC-ROC	AUC-PR
Logistic Regression	0.6839	0.5963
Random Forest	0.6954	0.6096
XGBoost (best)	0.7043	0.6247

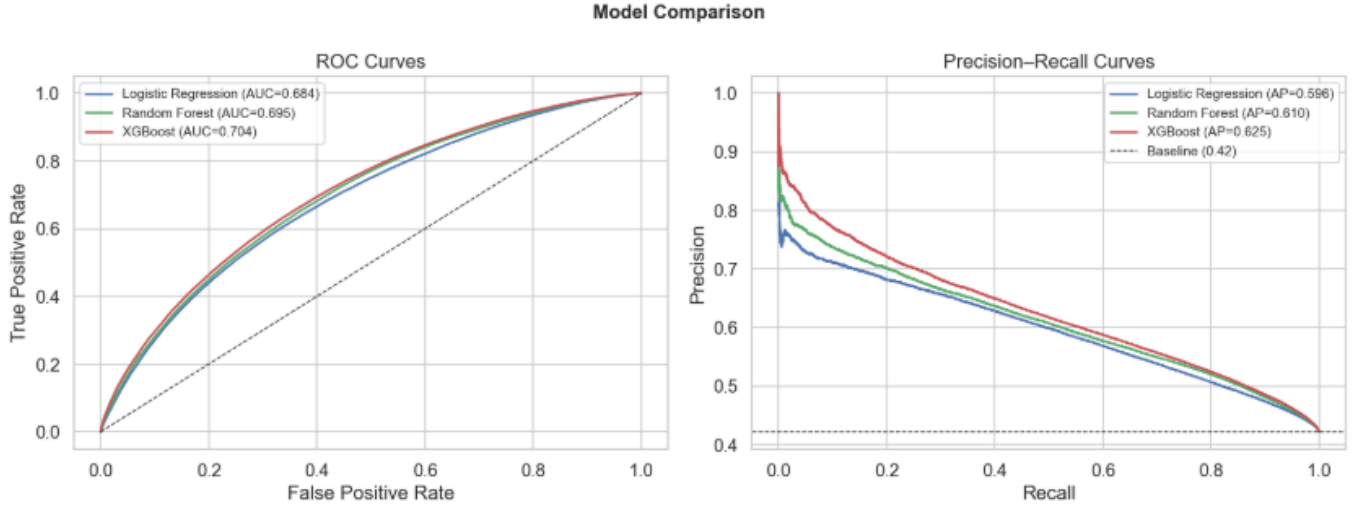


Fig. 3. ROC and Precision-Recall curves for all three classifiers



Fig. 4. SHAP feature importance: individual features (blue, left panel) versus neighborhood features (orange, right panel) by mean absolute SHAP value.

XGBoost achieves the highest performance on both metrics, with an AUC-ROC of 0.7043. [7]. The full classification report at threshold 0.5 is as follows: precision 0.69 and recall 0.74 for the no-recidivism class; precision 0.60 and recall 0.55 for the recidivism class; macro-averaged F1 of 0.64; overall accuracy of 0.66.

Figure 3 shows the ROC and Precision-Recall curves for all three classifiers.

B. SHAP Feature Importance and Neighborhood Effects

Figure 4 shows individual features versus neighborhood features ranked by mean absolute SHAP value.

The SHAP analysis is the analytical core of this study. Neighborhood features collectively account for **18.7%** of total mean absolute SHAP mass, while individual features account for 81.3%, as reported directly by the modeling notebook. Nearly one-fifth of the model's explanatory power comes from where a defendant lives, a contribution that COMPAS ignores.

C. Fairness Audit

Table VI reproduces the full fairness audit from the modeling notebook, now including AUC-ROC per group.

TABLE VI
FAIRNESS AUDIT BY RACE AND ETHNICITY (XGBOOST, $\tau = 0.5$)

Group	N	Actual Recidivism Rate	TPR	FPR	AUC-ROC
African American	58,470	0.464	0.573	0.282	0.707
Am. Indian / AK Native	11,897	0.558	0.680	0.429	0.683
Asian or Pacific Islander	2,546	0.397	0.634	0.295	0.722
Caucasian	179,717	0.402	0.534	0.253	0.704
Hispanic	18,920	0.396	0.470	0.232	0.671

The TPR gap between the highest group (American Indian or Alaskan Native, 0.680) and the lowest (Hispanic, 0.470) is 21 percentage points. The FPR gap is 20 percentage points. These disparities reflect the unequal base rates documented in Table I: American Indian or Alaskan Native defendants have an observed recidivism rate of 56.5%, compared to 38.8% for Hispanic defendants. Chouldechova [6] proved that these metric gaps are mathematically unavoidable when base rates differ; they cannot be eliminated by model choice alone. They are a consequence of structural inequality in the criminal justice system and in neighborhood conditions [18], not a correctable algorithmic failure. The contribution of this work is to show these figures transparently so communities can make informed deployment decisions.

Calibration across the deprivation spectrum is strong. The modeling notebook reports that mean predicted probabilities track mean actual rates within two percentage points in all five SEDI quintiles, as shown in Table VII.

TABLE VII
MODEL CALIBRATION BY SEDI QUINTILE

SEDI Quintile	Mean Predicted	Mean Actual	N (Test Set)
Q1	0.418	0.400	54,578
Q2	0.454	0.439	57,188
Q3	0.457	0.439	51,743
Q4	0.458	0.441	55,917
Q5	0.413	0.390	52,124

The model does not systematically over-predict or under-predict recidivism in the most deprived neighborhoods, which addresses a major concern with risk assessment tools.

VII. KEY FINDINGS AND INSIGHTS

Neighborhood features account for 18.7% of total SHAP explanation mass. SEDI and median household income appear in the top features globally, confirming that place-based disadvantage has independent predictive power above and beyond individual criminal history. This finding validates the motivation for constructing the SEDI composite rather than relying on individual census variables, which individually showed correlations below $|r| = 0.04$ with the target.

Both the SEDI quintile analysis and SHAP dependence plot show a quadratic pattern: recidivism peaks at moderate deprivation (Q2 through Q4) and is lower at both extremes. Linear models cannot capture this curvature, justifying gradient-boosted trees and retrospectively validating the use of a composite index rather than a linear deprivation term.

XGBoost achieves AUC-ROC of 0.7043, matching the published COMPAS range of 0.65 to 0.70 [7]. Competitive performance does not require proprietary data or non-disclosable algorithms.

Predicted probabilities track actual rates within two percentage points in all five SEDI quintiles, confirming the model does not systematically mispredict in the most or least deprived neighborhoods.

Individual predictors account for 81.3% of SHAP mass, removing neighborhood features would eliminate approximately 18.7% of explanatory power and would remove the only mechanism through which structural socioeconomic context enters the prediction.

VIII. CONCLUSION

This project demonstrated that a fully open-source, reproducible recidivism prediction system can match the predictive performance of COMPAS while simultaneously quantifying neighborhood effects and disclosing fairness metrics. The WCLD dataset, the SEDI composite index, and gradient-boosted trees provide a principled and auditable alternative to black-box commercial tools.

The SEDI engineering decision, justified on theoretical, statistical, and practical grounds shows that by collapsing collinear and nonlinearly acting census variables into a single composite index grounded in the public health and criminological literature, the analysis is able to attribute to neighborhood context the full share of predictive power.

The fairness audit reveals meaningful TPR and FPR disparities that are reported with full transparency so practitioners can weigh them in deployment decisions. These disparities are not a modeling failure but a reflection of structural inequality in base rates that must be acknowledged rather than hidden.

REFERENCES

- [1] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, “Machine bias: There is software used across the country to predict future criminals. And it is biased against Blacks,” *ProPublica*, May 2016. [Online]. Available: <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- [2] E. Ash, N. Goel, N. Li, C. Marangon, and P. Sun, *Wisconsin Circuit Court Longitudinal Data (WCLD)*, ETH Zurich and University of Oxford, 2021. [Online]. Available: <https://clezdata.github.io/wclld/>
- [3] T. Brennan, W. Dieterich, and B. Ehret, “Evaluating the predictive validity of the COMPAS risk and needs assessment system,” *Criminal Justice and Behavior*, vol. 36, no. 1, pp. 21–40, 2009.
- [4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [6] A. Chouldechova, “Fair prediction with disparate impact: A study of bias in recidivism prediction instruments,” *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.
- [7] J. Dressel and H. Farid, “The accuracy, fairness, and limits of predicting recidivism,” *Science Advances*, vol. 4, no. 1, p. eaao5580, 2018.
- [8] M. Hardt, E. Price, and N. Srebro, “Equality of opportunity in supervised learning,” in *Advances in Neural Information Processing Systems (NeurIPS 29)*, 2016, pp. 3315–3323.
- [9] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Trans. Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [10] A. J. H. Kind et al., “Neighborhood socioeconomic disadvantage and 30-day rehospitalization: A retrospective cohort study,” *Annals of Internal Medicine*, vol. 161, no. 11, pp. 765–774, 2014.
- [11] L. J. Krivo and R. D. Peterson, “Extremely disadvantaged neighborhoods and urban crime,” *Social Forces*, vol. 75, no. 2, pp. 619–648, 1996.
- [12] J. Larson, S. Mattu, L. Kirchner, and J. Angwin, “How we analyzed the COMPAS recidivism algorithm,” *ProPublica*, 2016. [Online]. Available: <https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>
- [13] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [14] A. R. Piquero, “The life-course view of criminal activity,” in *The Oxford Handbook of Criminological Theory*, F. T. Cullen and P. Wilcox, Eds., Oxford University Press, 2009.
- [15] S. F. Reardon and D. O’Sullivan, “Measures of spatial segregation,” *Sociological Methodology*, vol. 34, no. 1, pp. 121–162, 2011.
- [16] M. D. Reisig, K. Holtfreter, and M. Morash, “Assessing recidivism risk across female pathways to crime,” *Justice Quarterly*, vol. 23, no. 3, pp. 384–405, 2009.
- [17] R. J. Sampson, S. W. Raudenbush, and F. Earls, “Neighborhoods and violent crime: A multilevel study of collective efficacy,” *Science*, vol. 277, no. 5328, pp. 918–924, 1997.
- [18] R. J. Sampson, *Great American City: Chicago and the Enduring Neighborhood Effect*, University of Chicago Press, 2012.
- [19] C. R. Shaw and H. D. McKay, *Juvenile Delinquency and Urban Areas*, University of Chicago Press, 1942.
- [20] G. K. Singh, “Area deprivation and widening inequalities in US mortality, 1969–1998,” *American Journal of Public Health*, vol. 93, no. 7, pp. 1137–1143, 2003.
- [21] J. A. C. Sterne et al., “Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls,” *BMJ*, vol. 338, p. b2393, 2009.
- [22] P. Townsend, “Deprivation,” *Journal of Social Policy*, vol. 16, no. 2, pp. 125–146, 1987.
- [23] Wisconsin State Legislature, *Wisconsin Statutes, Chapter 939: Crimes General Provisions*, 2023. [Online]. Available: <https://docs.legis.wisconsin.gov/statutes/statutes/939>